

zadanie1_test

March 10, 2026

```
[1]: # 1. Ładowanie biblioteki Pandas
import pandas as pd
import numpy as np

# Zdefiniowanie nazwy wgranego pliku
file_name = 'IHME_GBD_2019_CHEWING_TOB_1990_2019_DATA_Y2021M05D27.CSV'
```

```
[2]: # 2. Tworzenie ramki danych ze słownika
dict_data = {
    'Imie': ['Anna', 'Jan', 'Piotr'],
    'Wiek': [28, 34, 29],
    'Miasto': ['Warszawa', 'Kraków', 'Poznań']
}
df_dict = pd.DataFrame(dict_data)
print("Ramka danych ze słownika:\n", df_dict, "\n")
```

Ramka danych ze słownika:

	Imie	Wiek	Miasto
0	Anna	28	Warszawa
1	Jan	34	Kraków
2	Piotr	29	Poznań

```
[3]: df = pd.read_csv(file_name, encoding='cp1250')
```

```
[4]: # 4. Tworzenie ramki danych z listy list
list_data = [
    ['Kasia', 22, 'Gdańsk'],
    ['Marek', 45, 'Wrocław'],
    ['Zofia', 31, 'Łódź']
]
df_list = pd.DataFrame(list_data, columns=['Imie', 'Wiek', 'Miasto'])
print("Ramka danych z listy list:\n", df_list, "\n")
```

Ramka danych z listy list:

	Imie	Wiek	Miasto
0	Kasia	22	Gdańsk
1	Marek	45	Wrocław

[5]: # 5. Transponowanie (zamiana kolumn i wierszy)

```
df_transposed = df_dict.T
print("Transponowana ramka ze słownika:\n", df_transposed, "\n")
```

Transponowana ramka ze słownika:

	0	1	2
Imie	Anna	Jan	Piotr
Wiek	28	34	29
Miasto	Warszawa	Kraków	Poznań

[6]: # 6. Wyświetlenie pierwszych 10 wierszy ramki danych

```
display(df.head(10))
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
5	5	Prevalence	1	Global	2	Female	
6	5	Prevalence	1	Global	1	Male	
7	5	Prevalence	1	Global	2	Female	
8	5	Prevalence	1	Global	1	Male	
9	5	Prevalence	1	Global	2	Female	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
5	8	15 to 19	332	Chewing tobacco	3	
6	8	15 to 19	332	Chewing tobacco	3	
7	8	15 to 19	332	Chewing tobacco	3	
8	8	15 to 19	332	Chewing tobacco	3	
9	8	15 to 19	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower
0	Rate	1990	0.038740	0.055586	0.027147
1	Rate	1990	0.011356	0.017594	0.007779
2	Rate	1991	0.039253	0.055838	0.027608
3	Rate	1991	0.011516	0.017807	0.007906
4	Rate	1992	0.039863	0.056448	0.027800
5	Rate	1992	0.011685	0.018425	0.007953
6	Rate	1993	0.040508	0.057025	0.028294

7	Rate	1993	0.011853	0.018675	0.008068
8	Rate	1994	0.041153	0.057284	0.028838
9	Rate	1994	0.012010	0.018950	0.008129

```
[7]: # 7. Wyświetlenie ostatnich 10 wierszy ramki danych
display(df.tail(10))
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
350540	5	Prevalence	522	Sudan	3	Both	
350541	5	Prevalence	522	Sudan	3	Both	
350542	5	Prevalence	522	Sudan	3	Both	
350543	5	Prevalence	522	Sudan	3	Both	
350544	5	Prevalence	522	Sudan	3	Both	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
350540	27	Age standardized	332	Chewing tobacco	3	
350541	27	Age standardized	332	Chewing tobacco	3	
350542	27	Age standardized	332	Chewing tobacco	3	
350543	27	Age standardized	332	Chewing tobacco	3	
350544	27	Age standardized	332	Chewing tobacco	3	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower
350540	Rate	2010	0.028506	0.034480	0.023124
350541	Rate	2011	0.028617	0.034465	0.023351
350542	Rate	2012	0.028790	0.034826	0.023438
350543	Rate	2013	0.028989	0.035133	0.023598
350544	Rate	2014	0.029232	0.035395	0.023858
350545	Rate	2015	0.029518	0.035685	0.024255
350546	Rate	2016	0.029855	0.036004	0.024500
350547	Rate	2017	0.029768	0.035934	0.024428
350548	Rate	2018	0.029760	0.035796	0.024462
350549	Rate	2019	0.029752	0.035857	0.024318

```
[8]: # 8. Wyświetlenie informacji o ramce danych
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 350550 entries, 0 to 350549
Data columns (total 16 columns):
```

#	Column	Non-Null Count	Dtype
0	measure_id	350550 non-null	int64
1	measure_name	350550 non-null	str
2	location_id	350550 non-null	int64
3	location_name	350550 non-null	str
4	sex_id	350550 non-null	int64
5	sex_name	350550 non-null	str
6	age_group_id	350550 non-null	int64
7	age_group_name	350550 non-null	str
8	rei_id	350550 non-null	int64
9	rei_name	350550 non-null	str
10	metric_id	350550 non-null	int64
11	metric_name	350550 non-null	str
12	year_id	350550 non-null	int64
13	val	350550 non-null	float64
14	upper	350550 non-null	float64
15	lower	350550 non-null	float64

dtypes: float64(3), int64(7), str(6)
memory usage: 42.8 MB

```
[9]: # 9. Wyświetlenie, ile wierszy i kolumn znajduje się w ramce danych
rows, cols = df.shape
print(f"\nLiczba wierszy: {rows}, Liczba kolumn: {cols}\n")
```

Liczba wierszy: 350550, Liczba kolumn: 16

```
[10]: # 10. Wyświetlenie informacji statystycznej o kolumnach liczbowych
display(df.describe())
```

	measure_id	location_id	sex_id	age_group_id	rei_id	\
count	350550.0	350550.000000	350550.000000	350550.000000	350550.0	
mean	5.0	135.639024	2.000000	29.421053	332.0	
std	0.0	98.136414	0.816498	48.993427	0.0	
min	5.0	1.000000	1.000000	8.000000	332.0	
25%	5.0	62.000000	1.000000	12.000000	332.0	
50%	5.0	122.000000	2.000000	17.000000	332.0	
75%	5.0	182.000000	3.000000	27.000000	332.0	
max	5.0	522.000000	3.000000	235.000000	332.0	

	metric_id	year_id	val	upper	lower
count	350550.0	350550.000000	350550.000000	350550.000000	350550.000000
mean	3.0	2004.500000	0.020179	0.032878	0.011704
std	0.0	8.655454	0.049594	0.070631	0.034127
min	3.0	1990.000000	0.001051	0.001597	0.000370
25%	3.0	1997.000000	0.002300	0.004438	0.001063
50%	3.0	2004.500000	0.004869	0.009064	0.002401

75%	3.0	2012.000000	0.014729	0.027427	0.007094
max	3.0	2019.000000	0.610180	0.773804	0.501187

```
[11]: # 11. Wyświetlenie informacji statystycznej o kolumnach kategoryzowanych (typu
      ↪ object/string)
      display(df.describe(include=['object', 'str']))
```

	measure_name	location_name	sex_name	age_group_name	rei_name	\
count	350550	350550	350550	350550	350550	
unique	1	205	3	19	1	
top	Prevalence	Global	Male	15 to 19	Chewing tobacco	
freq	350550	1710	116850	18450	350550	

	metric_name
count	350550
unique	1
top	Rate
freq	350550

```
[12]: # 12. Usunięcie brakujących wartości w ramce danych
      df_clean = df.dropna()
```

```
[13]: # 13. Przedstawienie wyboru wierszy i kolumn używając nazw oraz indeksów na
      ↪ różne sposoby
      # a) Wybór kolumny przez nazwę
      col_val = df['val']
      display(col_val)
      # b) Wybór wielu kolumn
      cols_subset = df[['location_name', 'year_id', 'val']]
      display(cols_subset)
      # c) Wybór po indeksach liczbowych (iloc) - wiersze od 0 do 4, kolumny od 0 do 2
      subset_iloc = df.iloc[0:5, 0:3]
      display(subset_iloc)
      # d) Wybór po nazwach kolumn i etykietach (loc)
      subset_loc = df.loc[0:4, ['location_name', 'val']]
      display(subset_loc)
```

0	0.038740
1	0.011356
2	0.039253
3	0.011516
4	0.039863
...	
350545	0.029518
350546	0.029855
350547	0.029768
350548	0.029760
350549	0.029752

Name: val, Length: 350550, dtype: float64

	location_name	year_id	val
0	Global	1990	0.038740
1	Global	1990	0.011356
2	Global	1991	0.039253
3	Global	1991	0.011516
4	Global	1992	0.039863
...	...	...	...
350545	Sudan	2015	0.029518
350546	Sudan	2016	0.029855
350547	Sudan	2017	0.029768
350548	Sudan	2018	0.029760
350549	Sudan	2019	0.029752

[350550 rows x 3 columns]

	measure_id	measure_name	location_id
0	5	Prevalence	1
1	5	Prevalence	1
2	5	Prevalence	1
3	5	Prevalence	1
4	5	Prevalence	1

	location_name	val
0	Global	0.038740
1	Global	0.011356
2	Global	0.039253
3	Global	0.011516
4	Global	0.039863

```
[14]: # 14. Wybór wierszy pod warunkiem odnośnie określonej wartości kolumny
condition1 = df[df['sex_name'] == 'Male']
display(condition1)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
2	5	Prevalence	1	Global	1	Male	
4	5	Prevalence	1	Global	1	Male	
6	5	Prevalence	1	Global	1	Male	
8	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350510	5	Prevalence	522	Sudan	1	Male	
350512	5	Prevalence	522	Sudan	1	Male	
350514	5	Prevalence	522	Sudan	1	Male	
350516	5	Prevalence	522	Sudan	1	Male	
350518	5	Prevalence	522	Sudan	1	Male	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	

4	8	15 to 19	332	Chewing tobacco	3
6	8	15 to 19	332	Chewing tobacco	3
8	8	15 to 19	332	Chewing tobacco	3
...	...	...	...	...	...
350510	27	Age standardized	332	Chewing tobacco	3
350512	27	Age standardized	332	Chewing tobacco	3
350514	27	Age standardized	332	Chewing tobacco	3
350516	27	Age standardized	332	Chewing tobacco	3
350518	27	Age standardized	332	Chewing tobacco	3

	metric_name	year_id	val	upper	lower
0	Rate	1990	0.038740	0.055586	0.027147
2	Rate	1991	0.039253	0.055838	0.027608
4	Rate	1992	0.039863	0.056448	0.027800
6	Rate	1993	0.040508	0.057025	0.028294
8	Rate	1994	0.041153	0.057284	0.028838
...	...	...	...	...	...
350510	Rate	2015	0.056920	0.069266	0.046399
350512	Rate	2016	0.057552	0.070067	0.046759
350514	Rate	2017	0.057348	0.069999	0.046643
350516	Rate	2018	0.057335	0.069531	0.046713
350518	Rate	2019	0.057328	0.069623	0.046442

[116850 rows x 16 columns]

```
[15]: # 15. Wybór wierszy pod warunkiem spełnienia kilku warunków jednocześnie
condition_multi = df[(df['year_id'] == 2019) & (df['sex_name'] == 'Female')]
display(condition_multi)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
59	5	Prevalence	1	Global	2	Female	
149	5	Prevalence	1	Global	2	Female	
239	5	Prevalence	1	Global	2	Female	
329	5	Prevalence	1	Global	2	Female	
419	5	Prevalence	1	Global	2	Female	
...	...	...	...	...	...	...	...
350159	5	Prevalence	522	Sudan	2	Female	
350249	5	Prevalence	522	Sudan	2	Female	
350339	5	Prevalence	522	Sudan	2	Female	
350429	5	Prevalence	522	Sudan	2	Female	
350519	5	Prevalence	522	Sudan	2	Female	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
59	8	15 to 19	332	Chewing tobacco	3	
149	9	20 to 24	332	Chewing tobacco	3	
239	10	25 to 29	332	Chewing tobacco	3	
329	11	30 to 34	332	Chewing tobacco	3	
419	12	35 to 39	332	Chewing tobacco	3	

...	...	...	...	...	...
350159	30	80 to 84	332	Chewing tobacco	3
350249	31	85 to 89	332	Chewing tobacco	3
350339	32	90 to 94	332	Chewing tobacco	3
350429	235	95 plus	332	Chewing tobacco	3
350519	27	Age standardized	332	Chewing tobacco	3

	metric_name	year_id	val	upper	lower
59	Rate	2019	0.012987	0.020436	0.008924
149	Rate	2019	0.014047	0.022941	0.008641
239	Rate	2019	0.019821	0.030219	0.013034
329	Rate	2019	0.023903	0.035252	0.016166
419	Rate	2019	0.029763	0.042043	0.020727

...	...	...	...	...	...
350159	Rate	2019	0.003342	0.007147	0.001225
350249	Rate	2019	0.003342	0.007147	0.001225
350339	Rate	2019	0.003342	0.007147	0.001225
350429	Rate	2019	0.003342	0.007147	0.001225
350519	Rate	2019	0.002105	0.002820	0.001568

[3895 rows x 16 columns]

```
[16]: # 16. Wybór wierszy, które zawierają w kolumnie kategoryzowanej określone słowo
contains_word = df[df['location_name'].str.contains('Poland', na=False)]
display(contains_word)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
70110	5	Prevalence	51	Poland	1	Male	
70111	5	Prevalence	51	Poland	2	Female	
70112	5	Prevalence	51	Poland	1	Male	
70113	5	Prevalence	51	Poland	2	Female	
70114	5	Prevalence	51	Poland	1	Male	
...	...	...	...	...	...	...	
71815	5	Prevalence	51	Poland	3	Both	
71816	5	Prevalence	51	Poland	3	Both	
71817	5	Prevalence	51	Poland	3	Both	
71818	5	Prevalence	51	Poland	3	Both	
71819	5	Prevalence	51	Poland	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
70110	8	15 to 19	332	Chewing tobacco	3	
70111	8	15 to 19	332	Chewing tobacco	3	
70112	8	15 to 19	332	Chewing tobacco	3	
70113	8	15 to 19	332	Chewing tobacco	3	
70114	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
71815	27	Age standardized	332	Chewing tobacco	3	
71816	27	Age standardized	332	Chewing tobacco	3	

71817	27	Age standardized	332	Chewing tobacco	3
71818	27	Age standardized	332	Chewing tobacco	3
71819	27	Age standardized	332	Chewing tobacco	3

	metric_name	year_id	val	upper	lower
70110	Rate	1990	0.006309	0.012000	0.002829
70111	Rate	1990	0.004738	0.010660	0.001733
70112	Rate	1991	0.006349	0.012210	0.002910
70113	Rate	1991	0.004763	0.010677	0.001790
70114	Rate	1992	0.006390	0.012312	0.002916
...	...	...	...	...	...
71815	Rate	2015	0.003120	0.003728	0.002606
71816	Rate	2016	0.003168	0.003803	0.002630
71817	Rate	2017	0.003217	0.003868	0.002667
71818	Rate	2018	0.003214	0.003860	0.002658
71819	Rate	2019	0.003210	0.003869	0.002654

[1710 rows x 16 columns]

```
[17]: # 17. Wybór wierszy, które nie zawierają w kolumnie kategoryzowanej określonego
      ↪ słowa
not_contains_word = df[~df['location_name'].str.contains('Global', na=False)]
display(not_contains_word)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
1710	5	Prevalence	6	China	1	Male	
1711	5	Prevalence	6	China	2	Female	
1712	5	Prevalence	6	China	1	Male	
1713	5	Prevalence	6	China	2	Female	
1714	5	Prevalence	6	China	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
1710	8	15 to 19	332	Chewing tobacco	3	
1711	8	15 to 19	332	Chewing tobacco	3	
1712	8	15 to 19	332	Chewing tobacco	3	
1713	8	15 to 19	332	Chewing tobacco	3	
1714	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	

350549	27	Age standardized	332	Chewing tobacco	3
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	metric_name	year_id	val	upper	lower
1710	Rate	1990	0.007743	0.014014	0.003551
1711	Rate	1990	0.004139	0.009927	0.001458
1712	Rate	1991	0.007733	0.014088	0.003545
1713	Rate	1991	0.004139	0.010053	0.001468
1714	Rate	1992	0.007723	0.014091	0.003576
...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255
350546	Rate	2016	0.029855	0.036004	0.024500
350547	Rate	2017	0.029768	0.035934	0.024428
350548	Rate	2018	0.029760	0.035796	0.024462
350549	Rate	2019	0.029752	0.035857	0.024318

[348840 rows x 16 columns]

```
[18]: # 18. Utworzenie kolumny na podstawie istniejących
df['val_percentage'] = df['val'] * 100
display(df)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

metric_name	year_id	val	upper	lower	val_percentage
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0	Rate	1990	0.038740	0.055586	0.027147	3.874015
1	Rate	1990	0.011356	0.017594	0.007779	1.135641
2	Rate	1991	0.039253	0.055838	0.027608	3.925338
3	Rate	1991	0.011516	0.017807	0.007906	1.151613
4	Rate	1992	0.039863	0.056448	0.027800	3.986274
...	...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255	2.951842
350546	Rate	2016	0.029855	0.036004	0.024500	2.985505
350547	Rate	2017	0.029768	0.035934	0.024428	2.976776
350548	Rate	2018	0.029760	0.035796	0.024462	2.975997
350549	Rate	2019	0.029752	0.035857	0.024318	2.975226

[350550 rows x 17 columns]

```
[19]: # 19. Usunięcie kolumny
df_dropped = df.drop(columns=['val_percentage'])
display(df_dropped)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower
0	Rate	1990	0.038740	0.055586	0.027147
1	Rate	1990	0.011356	0.017594	0.007779
2	Rate	1991	0.039253	0.055838	0.027608

3	Rate	1991	0.011516	0.017807	0.007906
4	Rate	1992	0.039863	0.056448	0.027800
...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255
350546	Rate	2016	0.029855	0.036004	0.024500
350547	Rate	2017	0.029768	0.035934	0.024428
350548	Rate	2018	0.029760	0.035796	0.024462
350549	Rate	2019	0.029752	0.035857	0.024318

[350550 rows x 6 columns]

```
[20]: # 20. Zmiana nazwy kolumny
df_renamed = df.rename(columns={'val': 'value', 'location_name': 'country'})
display(df_renamed)
```

	measure_id	measure_name	location_id	country	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

	metric_name	year_id	value	upper	lower	val_percentage
0	Rate	1990	0.038740	0.055586	0.027147	3.874015
1	Rate	1990	0.011356	0.017594	0.007779	1.135641
2	Rate	1991	0.039253	0.055838	0.027608	3.925338
3	Rate	1991	0.011516	0.017807	0.007906	1.151613
4	Rate	1992	0.039863	0.056448	0.027800	3.986274
...	...	...	...	...	...	...

350545	Rate	2015	0.029518	0.035685	0.024255	2.951842
350546	Rate	2016	0.029855	0.036004	0.024500	2.985505
350547	Rate	2017	0.029768	0.035934	0.024428	2.976776
350548	Rate	2018	0.029760	0.035796	0.024462	2.975997
350549	Rate	2019	0.029752	0.035857	0.024318	2.975226

[350550 rows x 17 columns]

```
[21]: # 21. Zachowanie ramki danych jako plik csv na komputerze
df_renamed.to_csv('przetworzone_dane.csv', index=False)
```

```
[22]: # 22. Wyświetlenie średniej, maksymalnej i minimalnej wartości z jednej kolumny
mean_val = df['val'].mean()
max_val = df['val'].max()
min_val = df['val'].min()
print(f"Średnia: {mean_val}, Max: {max_val}, Min: {min_val}\n")
```

Średnia: 0.020178816871399233, Max: 0.610180344, Min: 0.001050811

```
[23]: # 23. Wyświetlenie liczby wierszy
num_rows = len(df)
print(f"Liczba wierszy (len): {num_rows}\n")
```

Liczba wierszy (len): 350550

```
[24]: # 24. Wyświetlenie wartości unikatowych w kolumnie
unique_locations = df['location_name'].unique()
display(unique_locations)
```

```
<StringArray>
[
    'Global',
    'China',
    'Democratic People's Republic of Korea',
    'Taiwan (Province of China)',
    'Cambodia',
    'Indonesia',
    'Lao People's Democratic Republic',
    'Malaysia',
    'Maldives',
    'Myanmar',
    ...
    'Northern Mariana Islands',
    'Palau',
    'Puerto Rico',
    'Saint Kitts and Nevis',
    'San Marino',
    'Tokelau',
```

```

        'Tuvalu',
        'United States Virgin Islands',
        'South Sudan',
        'Sudan']

```

Length: 205, dtype: str

```

[25]: # 25. Wyświetlenie liczby rekordów odpowiadających danej wartości
value_counts_sex = df['sex_name'].value_counts()
display(value_counts_sex)

```

```

sex_name
Male      116850
Female    116850
Both      116850
Name: count, dtype: int64

```

```

[26]: # 26. Sortowanie wierszy według wartości kolumny (malejąco i rosnąco)
df_sorted_asc = df.sort_values(by='val', ascending=True)
df_sorted_desc = df.sort_values(by='val', ascending=False)

```

```

[27]: # 27. Wyświetlenie 10 największych (najmniejszych) wartości określonej kolumny
top_10 = df.nlargest(10, 'val')
bottom_10 = df.nsmallest(10, 'val')
display(top_10)
display(bottom_10)

```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
8127	5	Prevalence	10	Cambodia	2	Female	
8217	5	Prevalence	10	Cambodia	2	Female	
8307	5	Prevalence	10	Cambodia	2	Female	
8397	5	Prevalence	10	Cambodia	2	Female	
8125	5	Prevalence	10	Cambodia	2	Female	
8215	5	Prevalence	10	Cambodia	2	Female	
8305	5	Prevalence	10	Cambodia	2	Female	
8395	5	Prevalence	10	Cambodia	2	Female	
8129	5	Prevalence	10	Cambodia	2	Female	
8219	5	Prevalence	10	Cambodia	2	Female	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
8127	30	80 to 84	332	Chewing tobacco	3	
8217	31	85 to 89	332	Chewing tobacco	3	
8307	32	90 to 94	332	Chewing tobacco	3	
8397	235	95 plus	332	Chewing tobacco	3	
8125	30	80 to 84	332	Chewing tobacco	3	
8215	31	85 to 89	332	Chewing tobacco	3	
8305	32	90 to 94	332	Chewing tobacco	3	
8395	235	95 plus	332	Chewing tobacco	3	
8129	30	80 to 84	332	Chewing tobacco	3	
8219	31	85 to 89	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage
8127	Rate	2003	0.610180	0.708862	0.498634	61.018034
8217	Rate	2003	0.610180	0.708862	0.498634	61.018034
8307	Rate	2003	0.610180	0.708862	0.498634	61.018034
8397	Rate	2003	0.610180	0.708862	0.498634	61.018034
8125	Rate	2002	0.610154	0.716474	0.499911	61.015402
8215	Rate	2002	0.610154	0.716474	0.499911	61.015402
8305	Rate	2002	0.610154	0.716474	0.499911	61.015402
8395	Rate	2002	0.610154	0.716474	0.499911	61.015402
8129	Rate	2004	0.609981	0.706186	0.501187	60.998121
8219	Rate	2004	0.609981	0.706186	0.501187	60.998121

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
340439	5	Prevalence	396	San Marino	2	Female	
340437	5	Prevalence	396	San Marino	2	Female	
340429	5	Prevalence	396	San Marino	2	Female	
340431	5	Prevalence	396	San Marino	2	Female	
340435	5	Prevalence	396	San Marino	2	Female	
340433	5	Prevalence	396	San Marino	2	Female	
340427	5	Prevalence	396	San Marino	2	Female	
340425	5	Prevalence	396	San Marino	2	Female	
340423	5	Prevalence	396	San Marino	2	Female	
340421	5	Prevalence	396	San Marino	2	Female	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
340439	9	20 to 24	332	Chewing tobacco	3	
340437	9	20 to 24	332	Chewing tobacco	3	
340429	9	20 to 24	332	Chewing tobacco	3	
340431	9	20 to 24	332	Chewing tobacco	3	
340435	9	20 to 24	332	Chewing tobacco	3	
340433	9	20 to 24	332	Chewing tobacco	3	
340427	9	20 to 24	332	Chewing tobacco	3	
340425	9	20 to 24	332	Chewing tobacco	3	
340423	9	20 to 24	332	Chewing tobacco	3	
340421	9	20 to 24	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage
340439	Rate	2019	0.001051	0.002302	0.000383	0.105081
340437	Rate	2018	0.001051	0.002292	0.000387	0.105106
340429	Rate	2014	0.001051	0.002291	0.000418	0.105147
340431	Rate	2015	0.001053	0.002280	0.000399	0.105319
340435	Rate	2017	0.001054	0.002341	0.000394	0.105366
340433	Rate	2016	0.001054	0.002319	0.000403	0.105391
340427	Rate	2013	0.001055	0.002269	0.000411	0.105468
340425	Rate	2012	0.001057	0.002319	0.000406	0.105734
340423	Rate	2011	0.001060	0.002342	0.000408	0.106000
340421	Rate	2010	0.001063	0.002384	0.000392	0.106278

```
[28]: # 28. Wyświetlenie 10 największych wartości kolumny pod warunkiem określonej
      ↪wartości innej kolumny
      top_10_2019 = df[df['year_id'] == 2019].nlargest(10, 'val')
      display(top_10_2019)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
8159	5	Prevalence	10	Cambodia	2	Female	
8249	5	Prevalence	10	Cambodia	2	Female	
8339	5	Prevalence	10	Cambodia	2	Female	
8429	5	Prevalence	10	Cambodia	2	Female	
236488	5	Prevalence	164	Nepal	1	Male	
232169	5	Prevalence	161	Bangladesh	2	Female	
232259	5	Prevalence	161	Bangladesh	2	Female	
232349	5	Prevalence	161	Bangladesh	2	Female	
232439	5	Prevalence	161	Bangladesh	2	Female	
236398	5	Prevalence	164	Nepal	1	Male	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
8159	30	80 to 84	332	Chewing tobacco	3	
8249	31	85 to 89	332	Chewing tobacco	3	
8339	32	90 to 94	332	Chewing tobacco	3	
8429	235	95 plus	332	Chewing tobacco	3	
236488	13	40 to 44	332	Chewing tobacco	3	
232169	30	80 to 84	332	Chewing tobacco	3	
232259	31	85 to 89	332	Chewing tobacco	3	
232349	32	90 to 94	332	Chewing tobacco	3	
232439	235	95 plus	332	Chewing tobacco	3	
236398	12	35 to 39	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage
8159	Rate	2019	0.570629	0.722805	0.406286	57.062899
8249	Rate	2019	0.570629	0.722805	0.406286	57.062899
8339	Rate	2019	0.570629	0.722805	0.406286	57.062899
8429	Rate	2019	0.570629	0.722805	0.406286	57.062899
236488	Rate	2019	0.527289	0.633118	0.421249	52.728850
232169	Rate	2019	0.515467	0.618687	0.415773	51.546691
232259	Rate	2019	0.515467	0.618687	0.415773	51.546691
232349	Rate	2019	0.515467	0.618687	0.415773	51.546691
232439	Rate	2019	0.515467	0.618687	0.415773	51.546691
236398	Rate	2019	0.513928	0.622115	0.412132	51.392794

```
[29]: # 29. Grupowanie według kolumny kategoryzowanej, potem uśrednienie - MultiIndex
      ↪columns
      # Z uwagi na to, że mean() działa tylko na liczbach, wybieramy kolumny
      ↪numeryczne
      numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()
```



```

if 'year_id' in numeric_cols: numeric_cols.remove('year_id') # Rok to też
    ↳ liczba, ale chcemy średnią z danych
df_grouped_mean = df.groupby('location_name')[numeric_cols].mean()
display(df_grouped_mean.head())

```

	measure_id	location_id	sex_id	age_group_id	rei_id	\
location_name						
Afghanistan	5.0	160.0	2.0	29.421053	332.0	
Albania	5.0	43.0	2.0	29.421053	332.0	
Algeria	5.0	139.0	2.0	29.421053	332.0	
American Samoa	5.0	298.0	2.0	29.421053	332.0	
Andorra	5.0	74.0	2.0	29.421053	332.0	

	metric_id	val	upper	lower	val_percentage
location_name					
Afghanistan	3.0	0.036774	0.060089	0.020975	3.677381
Albania	3.0	0.005881	0.010670	0.003030	0.588069
Algeria	3.0	0.042593	0.073641	0.022522	4.259275
American Samoa	3.0	0.023810	0.041915	0.012453	2.381003
Andorra	3.0	0.001858	0.003450	0.000920	0.185760

```

[30]: # 30. Grupowanie z wieloma agregacjami na raz (średnia, liczba, mediana)
agg_funcs = {
    'val': 'mean',
    'upper': ['count', 'median']
}
df_grouped_agg = df.groupby('location_name').agg(agg_funcs)
display(df_grouped_agg.head())

```

	val	upper	
	mean	count	median
location_name			
Afghanistan	0.036774	1710	0.026186
Albania	0.005881	1710	0.006128
Algeria	0.042593	1710	0.074191
American Samoa	0.023810	1710	0.031380
Andorra	0.001858	1710	0.003368

```

[31]: # 31. Wyświetlenie nazw kolumn indeksu złożonego
print("Nazwy kolumn indeksu złożonego (MultiIndex):")
print(df_grouped_agg.columns)

```

```

Nazwy kolumn indeksu złożonego (MultiIndex):
MultiIndex([( 'val',  'mean'),
            ('upper', 'count'),
            ('upper', 'median')],
           )

```

```
[32]: # 32. Sortowanie kolumny w ramce z indeksem złożonym
# Sortujemy po (upper, median)
df_grouped_agg_sorted = df_grouped_agg.sort_values(by=('upper', 'median'),
↪ascending=False)
display(df_grouped_agg_sorted)
```

	val	upper	
	mean	count	median
location_name			
Nepal	0.257655	1710	0.393301
Bangladesh	0.308361	1710	0.383985
Bhutan	0.236710	1710	0.382318
India	0.218499	1710	0.310172
Palau	0.155873	1710	0.230229
...	...	...	...
Ireland	0.001443	1710	0.002759
Netherlands	0.001426	1710	0.002752
Spain	0.001424	1710	0.002742
France	0.001427	1710	0.002721
Portugal	0.001426	1710	0.002719

[205 rows x 3 columns]

```
[33]: # 33. Stworzenie tabeli przestawnej (pivot table)
pivot = pd.pivot_table(df, values='val', index=['location_name'],
↪columns=['year_id'], aggfunc='mean')
display(pivot)
```

year_id	1990	1991	1992	1993	\
location_name					
Afghanistan	0.033541	0.033696	0.033921	0.034097	
Albania	0.005795	0.005790	0.005785	0.005781	
Algeria	0.043953	0.043983	0.044003	0.044008	
American Samoa	0.023817	0.023812	0.023808	0.023802	
Andorra	0.001805	0.001806	0.001806	0.001807	
...	...	...	...	...	
Venezuela (Bolivarian Republic of)	0.012336	0.012334	0.012333	0.012330	
Viet Nam	0.023467	0.023431	0.023415	0.023408	
Yemen	0.108248	0.109595	0.110980	0.112388	
Zambia	0.011028	0.011035	0.011041	0.011047	
Zimbabwe	0.007388	0.007393	0.007398	0.007402	
year_id	1994	1995	1996	1997	\
location_name					
Afghanistan	0.034223	0.034356	0.034492	0.034551	
Albania	0.005778	0.005777	0.005776	0.005784	
Algeria	0.044011	0.044007	0.044000	0.043994	
American Samoa	0.023799	0.023795	0.023790	0.023786	

Andorra	0.001805	0.001804	0.001807	0.001809
...	...	...	...	...
Venezuela (Bolivarian Republic of)	0.012321	0.012311	0.012303	0.012297
Viet Nam	0.023397	0.023391	0.023388	0.023390
Yemen	0.113803	0.115222	0.116595	0.117946
Zambia	0.011054	0.011059	0.011061	0.011062
Zimbabwe	0.007407	0.007411	0.007420	0.007429

year_id	1998	1999	...	2010 \
location_name			...	
Afghanistan	0.034608	0.034679	...	0.038196
Albania	0.005790	0.005783	...	0.005963
Algeria	0.043977	0.043961	...	0.041578
American Samoa	0.023784	0.023777	...	0.023453
Andorra	0.001810	0.001810	...	0.001872
...	...	...	...	...
Venezuela (Bolivarian Republic of)	0.012292	0.012288	...	0.012306
Viet Nam	0.023395	0.023397	...	0.022982
Yemen	0.119187	0.120283	...	0.095544
Zambia	0.011065	0.011069	...	0.011417
Zimbabwe	0.007438	0.007444	...	0.007770

year_id	2011	2012	2013	2014 \
location_name				
Afghanistan	0.038576	0.039048	0.039553	0.040096
Albania	0.005983	0.006010	0.006037	0.006065
Algeria	0.040977	0.040432	0.039925	0.039615
American Samoa	0.023466	0.023692	0.023823	0.023988
Andorra	0.001884	0.001898	0.001915	0.001932
...	...	...	...	...
Venezuela (Bolivarian Republic of)	0.012312	0.012201	0.012086	0.012037
Viet Nam	0.022900	0.022823	0.022751	0.022697
Yemen	0.091861	0.088422	0.085207	0.083511
Zambia	0.011502	0.011541	0.011572	0.011717
Zimbabwe	0.007820	0.007989	0.008162	0.008337

year_id	2015	2016	2017	2018 \
location_name				
Afghanistan	0.040677	0.040325	0.040053	0.039984
Albania	0.006076	0.006080	0.006084	0.006076
Algeria	0.039339	0.039173	0.039010	0.039003
American Samoa	0.024060	0.024418	0.024416	0.024418
Andorra	0.001952	0.001970	0.001989	0.001998
...	...	...	...	...
Venezuela (Bolivarian Republic of)	0.011997	0.011990	0.011976	0.011972
Viet Nam	0.022625	0.022669	0.022715	0.022762
Yemen	0.082008	0.080679	0.079544	0.078674
Zambia	0.011858	0.012006	0.012170	0.012149

Zimbabwe	0.008477	0.008636	0.008796	0.008784
year_id	2019			
location_name				
Afghanistan	0.039886			
Albania	0.006067			
Algeria	0.038985			
American Samoa	0.024421			
Andorra	0.001998			
...	...			
Venezuela (Bolivarian Republic of)	0.011975			
Viet Nam	0.022805			
Yemen	0.077854			
Zambia	0.012139			
Zimbabwe	0.008773			

[205 rows x 30 columns]

```
[34]: # 34. Wyświetlenie indeksów i kolumn tabeli przestawnej
print("\nIndeksy tabeli przestawnej:", pivot.index)
print("\nKolumny tabeli przestawnej:", pivot.columns)
```

Indeksy tabeli przestawnej: Index(['Afghanistan', 'Albania', 'Algeria', 'American Samoa', 'Andorra', 'Angola', 'Antigua and Barbuda', 'Argentina', 'Armenia', 'Australia', ..., 'United States Virgin Islands', 'United States of America', 'Uruguay', 'Uzbekistan', 'Vanuatu', 'Venezuela (Bolivarian Republic of)', 'Viet Nam', 'Yemen', 'Zambia', 'Zimbabwe'], dtype='str', name='location_name', length=205)

Kolumny tabeli przestawnej: Index([1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019], dtype='int64', name='year_id')

```
[35]: # 35. Tworzenie indeksu złożonego dla tabeli przestawnej i wyświetlenie go
pivot_multi = pd.pivot_table(df, values='val', index=['location_name', 'sex_name'], columns=['year_id'], aggfunc='mean')
display(pivot_multi.head())
```

year_id		1990	1991	1992	1993	1994 \
location_name	sex_name					
Afghanistan	Both	0.032164	0.032468	0.032946	0.033264	0.033423
	Female	0.005649	0.005646	0.005642	0.005640	0.005639
	Male	0.062808	0.062973	0.063175	0.063388	0.063608
Albania	Both	0.005851	0.005838	0.005825	0.005812	0.005800

	Female	0.003087	0.003086	0.003085	0.003084	0.003084	
year_id		1995	1996	1997	1998	1999	...
location_name	sex_name						...
Afghanistan	Both	0.033584	0.033740	0.033816	0.033861	0.033922	...
	Female	0.005639	0.005640	0.005570	0.005507	0.005450	...
	Male	0.063846	0.064096	0.064266	0.064455	0.064663	...
Albania	Both	0.005790	0.005781	0.005789	0.005797	0.005773	...
	Female	0.003083	0.003083	0.003085	0.003085	0.003083	...

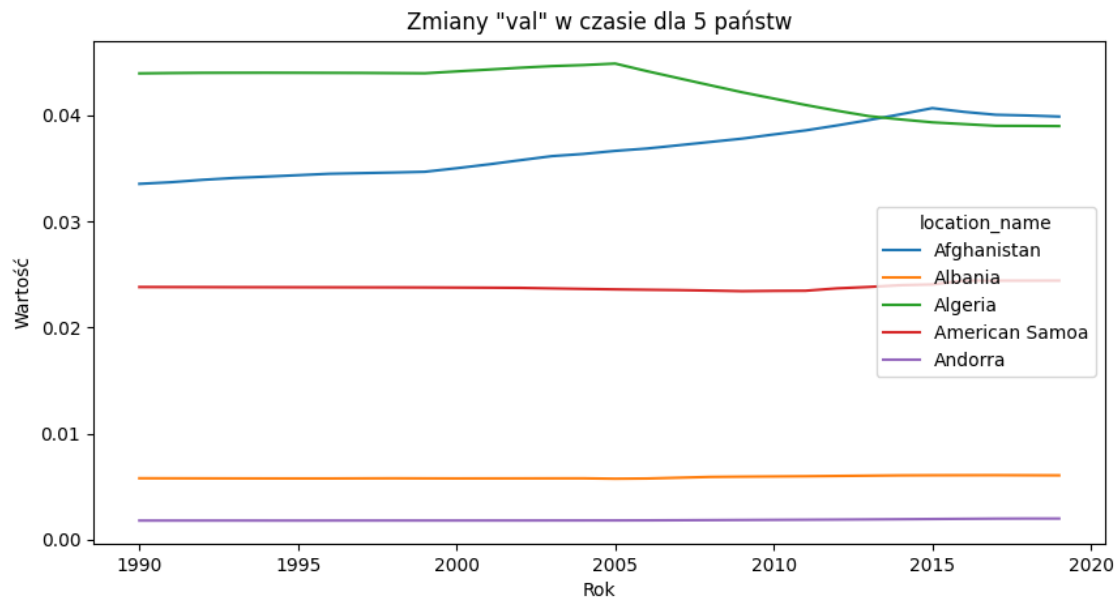
year_id		2010	2011	2012	2013	2014	...
location_name	sex_name						...
Afghanistan	Both	0.038310	0.038805	0.039383	0.039976	0.040568	
	Female	0.004710	0.004636	0.004573	0.004517	0.004428	
	Male	0.071569	0.072288	0.073188	0.074167	0.075293	
Albania	Both	0.005924	0.005944	0.005969	0.005995	0.006022	
	Female	0.003138	0.003163	0.003191	0.003220	0.003250	

year_id		2015	2016	2017	2018	2019	
location_name	sex_name						
Afghanistan	Both	0.041183	0.040883	0.040657	0.040620	0.040548	
	Female	0.004351	0.004285	0.004234	0.004232	0.004232	
	Male	0.076496	0.075808	0.075268	0.075099	0.074878	
Albania	Both	0.006034	0.006041	0.006047	0.006042	0.006036	
	Female	0.003264	0.003274	0.003284	0.003280	0.003278	

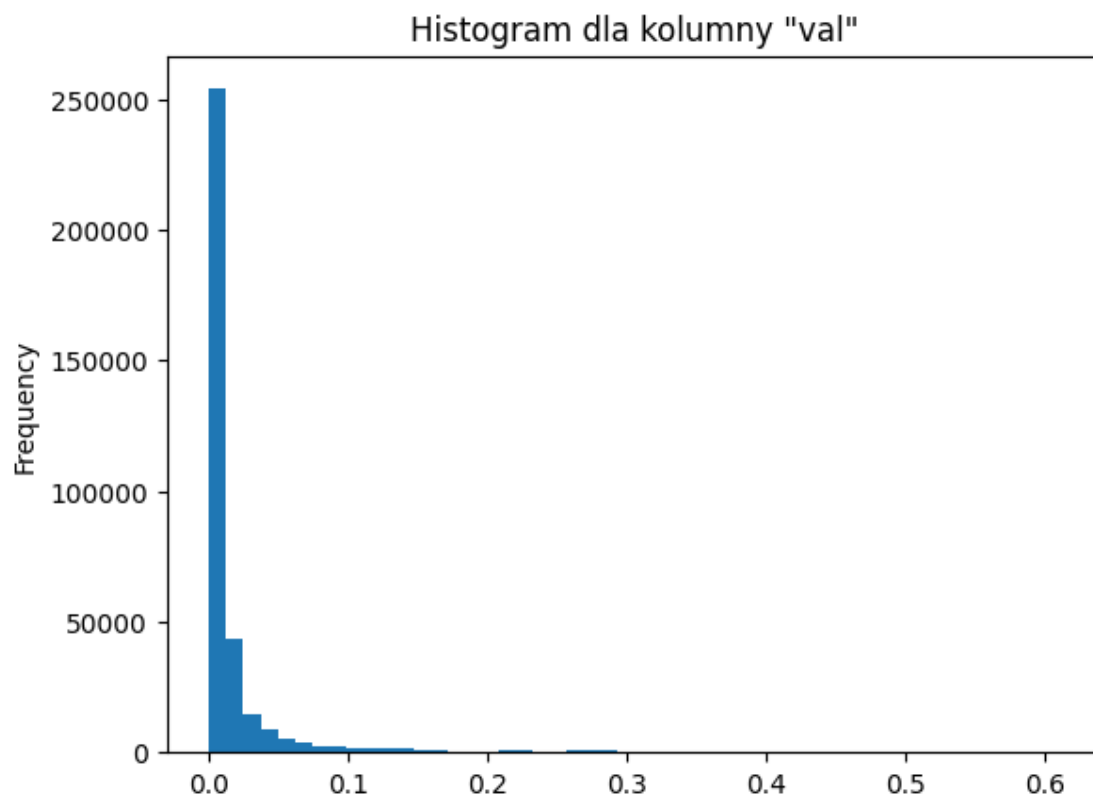
[5 rows x 30 columns]

```
[36]: # 36. Import modułu pyplot z biblioteki matplotlib
import matplotlib.pyplot as plt
```

```
[37]: # 38. Wyświetlenie wykresu na podstawie tabeli przestawnej
# Wykres dla pierwszych 5 krajów
pivot.head(5).T.plot(kind='line', title='Zmiany "val" w czasie dla 5 państw',
    figsize=(10, 5))
plt.ylabel('Wartość')
plt.xlabel('Rok')
plt.show()
```



```
[38]: # 39. Narysowanie histogramu na podstawie wartości kolumny
df['val'].plot(kind='hist', bins=50, title='Histogram dla kolumny "val"')
plt.show()
```



```
[39]: # 40. Przedstawienie łączenia ramek danych za pomocą metod merge i concat
df_part1 = df.iloc[:100]
df_part2 = df.iloc[100:200]
# Concat - sklejanie jeden pod drugim
df_concat = pd.concat([df_part1, df_part2])

# Merge - łączenie na podstawie wspólnego klucza (np. location_id)
# Tworzymy dodatkową tabelę (słownik) państw do złączenia
regions = pd.DataFrame({'location_id': [1, 522], 'Region_Type': ['Global', 'Country']})
df_merged = pd.merge(df, regions, on='location_id', how='left')
display(df_merged)
display(df_concat)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage	\
0	Rate	1990	0.038740	0.055586	0.027147	3.874015	
1	Rate	1990	0.011356	0.017594	0.007779	1.135641	
2	Rate	1991	0.039253	0.055838	0.027608	3.925338	
3	Rate	1991	0.011516	0.017807	0.007906	1.151613	

4	Rate	1992	0.039863	0.056448	0.027800	3.986274
...	...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255	2.951842
350546	Rate	2016	0.029855	0.036004	0.024500	2.985505
350547	Rate	2017	0.029768	0.035934	0.024428	2.976776
350548	Rate	2018	0.029760	0.035796	0.024462	2.975997
350549	Rate	2019	0.029752	0.035857	0.024318	2.975226

	Region_Type
0	Global
1	Global
2	Global
3	Global
4	Global
...	...
350545	Country
350546	Country
350547	Country
350548	Country
350549	Country

[350550 rows x 18 columns]

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
..	...	...	...	...	...	...	
195	5	Prevalence	1	Global	2	Female	
196	5	Prevalence	1	Global	1	Male	
197	5	Prevalence	1	Global	2	Female	
198	5	Prevalence	1	Global	1	Male	
199	5	Prevalence	1	Global	2	Female	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
..	...	...	...	...	...	
195	10	25 to 29	332	Chewing tobacco	3	
196	10	25 to 29	332	Chewing tobacco	3	
197	10	25 to 29	332	Chewing tobacco	3	
198	10	25 to 29	332	Chewing tobacco	3	
199	10	25 to 29	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage
0	Rate	1990	0.038740	0.055586	0.027147	3.874015
1	Rate	1990	0.011356	0.017594	0.007779	1.135641
2	Rate	1991	0.039253	0.055838	0.027608	3.925338
3	Rate	1991	0.011516	0.017807	0.007906	1.151613
4	Rate	1992	0.039863	0.056448	0.027800	3.986274
..	...	...	...	...	...	...
195	Rate	1997	0.017918	0.027358	0.011525	1.791802
196	Rate	1998	0.069306	0.087765	0.053291	6.930626
197	Rate	1998	0.018165	0.027326	0.011713	1.816517
198	Rate	1999	0.070359	0.088517	0.054634	7.035858
199	Rate	1999	0.018439	0.027995	0.012031	1.843930

[200 rows x 17 columns]

```
[40]: # 41. Pokazanie dodawania nowych kolumn za pomocą operacji matematycznych
df['rozstep'] = df['upper'] - df['lower']
display(df)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	
3	8	15 to 19	332	Chewing tobacco	3	
4	8	15 to 19	332	Chewing tobacco	3	
...	...	...	...	...	...	
350545	27	Age standardized	332	Chewing tobacco	3	
350546	27	Age standardized	332	Chewing tobacco	3	
350547	27	Age standardized	332	Chewing tobacco	3	
350548	27	Age standardized	332	Chewing tobacco	3	
350549	27	Age standardized	332	Chewing tobacco	3	

	metric_name	year_id	val	upper	lower	val_percentage	\
0	Rate	1990	0.038740	0.055586	0.027147	3.874015	

1	Rate	1990	0.011356	0.017594	0.007779	1.135641
2	Rate	1991	0.039253	0.055838	0.027608	3.925338
3	Rate	1991	0.011516	0.017807	0.007906	1.151613
4	Rate	1992	0.039863	0.056448	0.027800	3.986274
...	...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255	2.951842
350546	Rate	2016	0.029855	0.036004	0.024500	2.985505
350547	Rate	2017	0.029768	0.035934	0.024428	2.976776
350548	Rate	2018	0.029760	0.035796	0.024462	2.975997
350549	Rate	2019	0.029752	0.035857	0.024318	2.975226

	rozstep
0	0.028439
1	0.009816
2	0.028230
3	0.009901
4	0.028648
...	...
350545	0.011430
350546	0.011504
350547	0.011506
350548	0.011334
350549	0.011539

[350550 rows x 18 columns]

```
[41]: # 42. Pokazanie dodawania nowych kolumn za pomocą funkcji lambda
# Przykładowo: Sprawdzenie, czy wartość 'val' jest wyższa niż 0.02
df['is_high'] = df['val'].apply(lambda x: 'Yes' if x > 0.02 else 'No')
display(df)
```

	measure_id	measure_name	location_id	location_name	sex_id	sex_name	\
0	5	Prevalence	1	Global	1	Male	
1	5	Prevalence	1	Global	2	Female	
2	5	Prevalence	1	Global	1	Male	
3	5	Prevalence	1	Global	2	Female	
4	5	Prevalence	1	Global	1	Male	
...	...	...	...	...	...	...	
350545	5	Prevalence	522	Sudan	3	Both	
350546	5	Prevalence	522	Sudan	3	Both	
350547	5	Prevalence	522	Sudan	3	Both	
350548	5	Prevalence	522	Sudan	3	Both	
350549	5	Prevalence	522	Sudan	3	Both	

	age_group_id	age_group_name	rei_id	rei_name	metric_id	\
0	8	15 to 19	332	Chewing tobacco	3	
1	8	15 to 19	332	Chewing tobacco	3	
2	8	15 to 19	332	Chewing tobacco	3	

3	8	15 to 19	332	Chewing tobacco	3
4	8	15 to 19	332	Chewing tobacco	3
...	...	...	...	...	...
350545	27	Age standardized	332	Chewing tobacco	3
350546	27	Age standardized	332	Chewing tobacco	3
350547	27	Age standardized	332	Chewing tobacco	3
350548	27	Age standardized	332	Chewing tobacco	3
350549	27	Age standardized	332	Chewing tobacco	3

	metric_name	year_id	val	upper	lower	val_percentage	\
0	Rate	1990	0.038740	0.055586	0.027147	3.874015	
1	Rate	1990	0.011356	0.017594	0.007779	1.135641	
2	Rate	1991	0.039253	0.055838	0.027608	3.925338	
3	Rate	1991	0.011516	0.017807	0.007906	1.151613	
4	Rate	1992	0.039863	0.056448	0.027800	3.986274	
...	...	...	...	...	...	...	...
350545	Rate	2015	0.029518	0.035685	0.024255	2.951842	
350546	Rate	2016	0.029855	0.036004	0.024500	2.985505	
350547	Rate	2017	0.029768	0.035934	0.024428	2.976776	
350548	Rate	2018	0.029760	0.035796	0.024462	2.975997	
350549	Rate	2019	0.029752	0.035857	0.024318	2.975226	

	rozstep	is_high
0	0.028439	Yes
1	0.009816	No
2	0.028230	Yes
3	0.009901	No
4	0.028648	Yes
...	...	...
350545	0.011430	Yes
350546	0.011504	Yes
350547	0.011506	Yes
350548	0.011334	Yes
350549	0.011539	Yes

[350550 rows x 19 columns]

```
[42]: # 43. Przedstawienie możliwości pracy z dużymi plikami przy użyciu argumentu
      ↪ chunksize
chunk_iterator = pd.read_csv(file_name, chunksize=1000, encoding='cp1250')
# Pętla po "kawałkach"
total_rows = 0
for chunk in chunk_iterator:
    # W każdej pętli ramka danych 'chunk' zawiera maksymalnie 1000 wierszy
    total_rows += len(chunk)

print(f"\nPrzetworzono łącznie {total_rows} wierszy przy użyciu chunksize.")
```

Przetworzono łącznie 350550 wierszy przy użyciu chunksize.

[]: